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Hemostatic Product Strategy for Non-Compressible and Junctional Hemorrhage Control

Executive summary

A single "universal" hemostatic product for groin, axilla, neck, pelvic, and internal bleeding is not the strongest development strategy. The anatomy, deployment conditions, and regulatory categories are too different. Current evidence and cleared products support a **two-platform strategy** instead: a **temporary, pressure-generating, retrievable junctional device** for deep groin/axillary wounds, and a **separate absorbable internal-use sealant or patch platform** for operative or catheter-directed control of pelvic and intra-abdominal bleeding. Existing products illustrate why: XSTAT is effective for deep groin/axillary junctional wounds but is explicitly not indicated for the abdomen, retroperitoneum, or tissues above the clavicle, while internal surgical products such as TachoSil, EVARREST, TISSEEL, VISTASEAL, FLOSEAL, and ETHIZIA are intended for controlled operative use rather than field deployment. [1]

For a **first product**, the best-risk-adjusted target is **deep junctional groin/axillary hemorrhage not amenable to tourniquet use**, where a syringe-deployed, radiopaque, blood-triggered **shape-memory sponge or cryogel** has the clearest clinical logic, the strongest military/prehospital precedent, and a defined U.S. regulatory path via the XSTAT device class. The product should be **temporary, pressure-limited, highly radiopaque, easy to count and retrieve**, and usable by minimally trained responders under low-light, high-stress conditions. [2]

For **internal and pelvic bleeding**, the evidence favors a different architecture: **wet-tissue adhesive hydrogels, fibrin or synthetic hemostatic patches, flowable gelatin-thrombin matrices, or catheter-deliverable embolic materials**. These products work because they seal or accelerate clot formation on organ surfaces or within defined surgical fields, but they are much less suitable for field use in the neck or uncontrolled torso cavities. Pelvic arterial bleeding in particular is commonly treated with **transcatheter arterial embolization**, which means a future pelvic/internal program should be designed either for **operating room use** or for **interventional radiology delivery**, not as a free-expanding prehospital cavity filler. [3]

From a materials standpoint, the most promising toolbox combines **mechanical expansion**, **procoagulant activity**, and **wet-tissue adhesion** rather than relying on a single mechanism. Practical candidates include **cellulose/chitosan/gelatin cryogels**, **recombinant collagen-PEG-dopamine cryogels**, **kaolin- or thrombin-decorated sponges**, **PEG-based adhesive hydrogels**, **GelMA/quaternary chitosan/calcium systems**, **alginate-carboxymethyl chitosan foams or gels**, and **catechol-functionalized bioadhesives**. Local **tranexamic acid** delivery is attractive as an adjunct, especially in deep or coagulopathic bleeding models, but the evidence remains predominantly preclinical and it should not be the sole hemostatic mechanism. [4]

The current commercial landscape is crowded in **external compressible bleeding** and in **operative internal adjunctive hemostasis**, but it remains relatively sparse in **prehospital non-compressible torso**

and pelvic bleeding. That whitespace exists for a reason: the hardest translational problems are **embolization risk, airway/vascular compression, removal or bioresorption, infection, and damage from swelling or pressurization**. ResQFoam and related abdominal foam concepts show the opportunity but also the challenge, with survival gains in swine accompanied by increased intra-abdominal pressure and bowel injury at higher doses. [5]

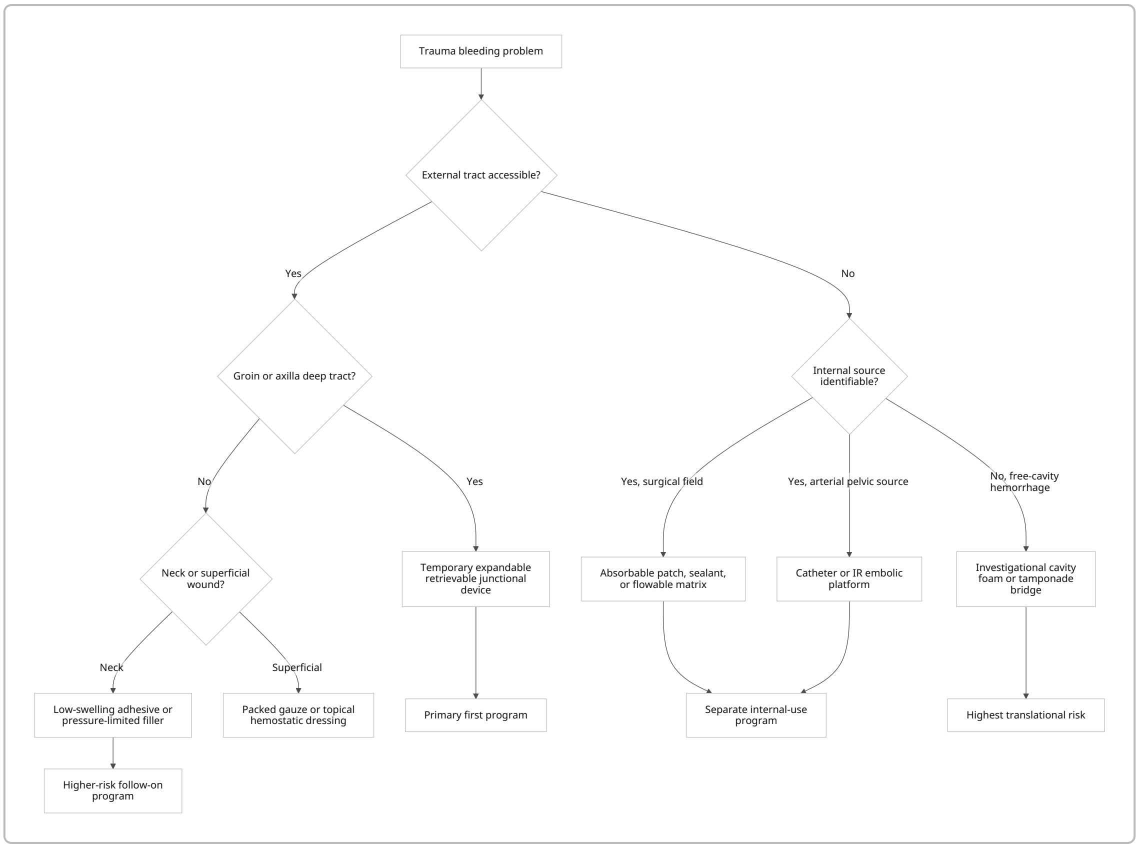
Clinical need and target use-cases

Uncontrolled hemorrhage remains one of the leading preventable causes of trauma death, and **junctional** plus **non-compressible torso** bleeding remain especially difficult because the usual tools—manual pressure and limb tourniquets—either work poorly or do not apply. The Joint Trauma System's damage-control guidance explicitly distinguishes superficial wounds manageable with hemostatic dressings from junctional bleeding that may require XSTAT or junctional tourniquets, while truncal internal hemorrhage remains an area of intensive research. Reviews of non-compressible torso hemorrhage describe the same problem from the biomaterials side: organ, pelvic, and cavity bleeding are not amenable to conventional external compression. [6]

The clinical use-cases divide into four distinct engineering problems. **Groin and axilla** are deep, narrow, and often irregular tract wounds; products must move quickly down the wound tract and then expand or seal without relying on perfect wound visualization. **Neck wounds** are even more constrained because any uncontrolled swelling risks compression of the airway, carotid artery, jugular vein, or cranial nerves; it is notable that XSTAT is not indicated for tissues above the clavicle. **Pelvic internal bleeding** is often retroperitoneal and arterial or venous rather than directly accessible through an external tract, which is why embolization is a major standard approach. **Intra-abdominal bleeding** further divides into surface bleeding that a patch or sealant can contact versus free-cavity or hidden-source bleeding that might require tamponade, embolization, or definitive surgery. [7]

These differences matter because they determine what "success" means. In groin or axilla, success is rapid deployment into a narrow tract by a responder with limited training. In the neck, success also requires **strict control of expansion and pressure**. In pelvic and abdominal bleeding, success may instead mean **surgeon- or catheter-delivered control** of an internal source until definitive repair. The evidence base therefore argues against a single product claim spanning all sites on day one; a staged claim strategy is more credible clinically and regulatorily. [8]

A realistic target product profile for a first indication would therefore be: **deep junctional hemorrhage in groin/axilla, non-compressible, not amenable to tourniquet use, field-deployable, temporary, retrievable, and effective under coagulopathic conditions**. Neck and internal/pelvic use should be treated either as later indications or as separate programs with different delivery systems and safety controls. [9]



Hemostatic mechanisms and candidate formulations

The current product landscape shows that no single hemostatic mechanism is sufficient across all non-compressible settings. Existing products work through combinations of **physical barrier/tamponade**, **fluid absorption and concentration of cellular/clotting components**, **procoagulant signaling**, and **wet-tissue adhesion**. XSTAT exemplifies the mechanical route: compressed radiopaque cellulose sponges expand upon contact with blood to fill a wound cavity and create localized pressure. Kaolin dressings such as QuikClot use a mineral hemostat bound to gauze, while chitosan systems such as Celox and ChitoGauze form a gel-like barrier and concentrate platelets or cellular components at the wound. Fibrin sealants and fibrin patches reproduce the final steps of the coagulation cascade by combining thrombin and fibrinogen at the bleeding surface. [10]

Procoagulant agents

The best-supported procoagulant actives for a trauma program are **kaolin**, **chitosan**, **thrombin/ fibrinogen**, and possibly **calcium-containing** adjuncts. Kaolin is already embedded in QuikClot dressings and promotes clotting on contact with blood. Chitosan remains attractive because it appears in multiple 510(k)-cleared products and in newer gels and cryogels. Thrombin and fibrinogen are clinically powerful but pull the program toward more complex surgical or biologic regulatory pathways. Calcium-ion-containing systems, including GelMA/quaternary chitosan/calcium hydrogels and $CaCO_3$-based TXA spheres, can reinforce coagulation-favorable local environments. [11]

Mechanical expansion and cavity filling

Mechanical expansion is the most intuitive mechanism for **deep junctional wounds** because it does not require pristine tissue surfaces or perfect lesion visualization. Published cryogel and sponge systems use **blood-triggered or liquid-triggered shape recovery** to re-expand after syringe delivery. Recent examples include recombinant collagen/PEG/dopamine cryogels with rapid shape recovery, chitosan-based cryogels with high absorbency and fast recovery, and thrombin-decorated rapidly expandable cryogels that achieved rapid hemostasis and high survival in a swine junctional model. This category aligns best with groin and axillary wounds. [12]

Adhesive chemistries

Wet-tissue adhesion is the critical mechanism for **visceral organ surfaces, irregular cavities, and pulsatile bleeding interfaces**. The literature highlights several adhesive families: **catechol-based systems** inspired by mussel chemistry, **PEG-based covalent adhesives**, **methacrylated gelatin and alginate systems**, and **Schiff-base aldehyde/amine networks**. Strongly adhesive hydrogels in this class have been reported to seal arterial and cardiac bleeding, including published burst-pressure performance around 290 mmHg in one notable system. Liquid-infused bioadhesives have likewise been shown to halt non-compressible hemorrhage in animal models. [13]

In situ gelling polymers, foams, and hydrogels

In situ gels and foams solve the delivery problem differently: they are injected or sprayed as liquids or pastes and then **solidify, gel, or expand** after contact with the wound or after external triggering. Promising examples include **PEG-based injectable hydrogels with on-demand dissolution**, **thermogelling chitosan-based bioadhesive hydrogels**, **GelMA/quaternary chitosan/calcium blue-light-triggered adhesives**, and **alginate/carboxymethyl-chitosan-based foams** intended for abdominal hemorrhage. This family is particularly relevant for internal bleeding and for surgeon- or catheter-delivered systems, but the main challenge is balancing rapid solidification against embolization risk and safe removal or resorption. [14]

Tranexamic acid delivery

Local **tranexamic acid** is appealing because it addresses fibrinolysis, which can be a major trauma problem, but the strongest case is as an **adjunct** layered onto a mechanical or adhesive platform. TXA-enhanced chitosan or alginate hydrogels, TXA-induced fast-gelling PDMAA/carboxymethyl-chitosan systems, and TXA-loaded self-propelled hydrogel spheres have shown improved hemostasis in deep or non-compressible bleeding models. The caution is that these are still largely preclinical material platforms, not yet an established commercial standard for traumatic internal hemorrhage control. [15]

Candidate material and formulation set

Platform	Representative composition	Crosslinking or setting logic	Best-fit use-case	Main trade-off
Expandable retrievable sponge	Cellulose or recombinant collagen/ PEG/dopamine cryogel with radiopaque marker; optional kaolin or thrombin surface loading [16]	Blood-triggered expansion or shape-memory recovery	Deep groin/ axillary tract wounds	Retrieval burden; pressure injury risk if expansion is uncontrolled
Chitosan gauze or gel	Chitosan granules on gauze or in alginate gel [17]	Physical barrier plus platelet concentration and gel-like plug formation	External moderate-severe bleeding, irregular superficial cavities	External use only in current clearances
Kaolin gauze	Kaolin bound to medical gauze [18]	Mineral hemostat on packed gauze	Field wound packing	Requires firm packing and pressure; limited in non-packable tracts
PEG adhesive hydrogel	Injectable PEG-based covalent hydrogel with optional on-demand dissolution [19]	Rapid covalent gelation	Internal surfaces, surgeon-delivered sealant	Mixing, embolization control, and removal strategy matter
GelMA/ chitosan adhesive	GelMA + quaternary chitosan + Ca^{2+} [20]	Blue-light-triggered crosslinking plus hydrogen bonding	Operative field, visible bleeding surfaces	Light source requirement limits field use
Aldehyde-amine hydrogel	Oxidized CMC + gelatin [21]	Schiff-base plus amide network	Injectable internal sealant	Cytotoxicity and degradation chemistry must be tightly controlled
Alginate-based system	Alginate or dopamine-modified alginate + GelMA; optional TXA [22]	Ionic or photo-crosslinking	Sprayable or injectable internal-use hydrogel	Swelling and washout must be controlled

Platform	Representative composition	Crosslinking or setting logic	Best-fit use-case	Main trade-off
Fibrin patch or sealant	Human fibrinogen + human thrombin on collagen patch or in two-component liquid [23]	Final common coagulation pathway recreated at wound surface	Operative hemostasis on organ surfaces	Cost, supply chain, biologic regulation, intravascular thrombosis risk

Device concepts and deployment pathways

The most credible **junctional product concept** is a **self-expanding, pressure-generating, retrievable sponge/cryogel cartridge**. This is the concept space validated by XSTAT and by newer thrombin-decorated cryogels in swine. It is well suited to groin and axilla because it can be delivered down a narrow tract, then re-expand to tamponade the wound while also carrying procoagulant chemistry. To improve on XSTAT, the next-generation design should ideally add **more conformal expansion, better wound-wall adhesion, and more forgiving retrievability**, while keeping **radiopacity and countability**. [24]

The best **second concept** for internal use is a **surgeon-deliverable absorbable patch or injectable adhesive hydrogel** rather than a field foam. Internal patches now have strong commercial and clinical validation: TachoSil and EVARREST are fibrin patches, and ETHIZIA is a newer synthetic-polymer/gelatin patch cleared for open liver surgery with faster median time to hemostasis than TachoSil in its pivotal comparison. This category is strongest when bleeding is **visible** and the user can apply compression or hold the patch in place for seconds to minutes. [25]

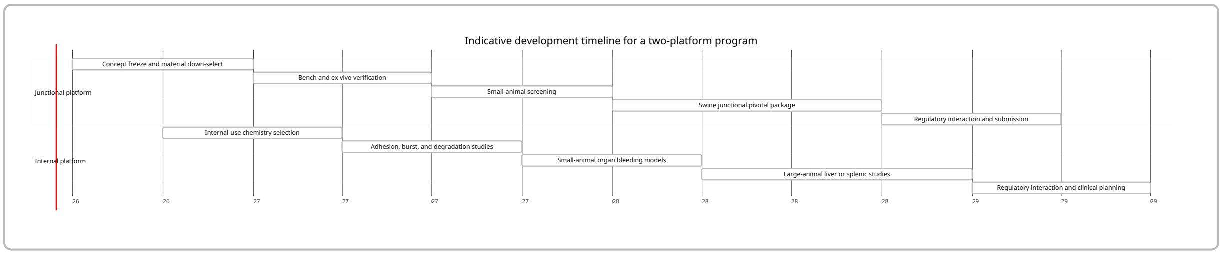
A third concept is a **flowable or sprayable sealant** for hard-to-reach surfaces. TISSEEL and VISTASEAL show how two-component fibrin sealants are delivered in surgery, while TRAUMAGEL shows that a chitosan-alginate gel can be cleared for external moderate-to-severe bleeding. For internal trauma, a sprayable or flowable sealant would need: rapid set, strong wet adhesion, a clear visual field, no intravascular migration, and highly controlled gas-assisted delivery if sprayed. [26]

A fourth concept is an **intra-abdominal expansile foam** for non-compressible torso hemorrhage. This remains strategically important but translationally difficult. ResQFoam and related self-expanding foam systems improved survival and reduced hemorrhage in severe swine injury models and have progressed to early clinical evaluation, but higher-dose abdominal foam was also associated with increased intra-abdominal pressure and bowel injury. This space remains high-reward, high-risk, and is a poor first indication unless the company is prepared for a longer and more complex clinical program. [5]

For **pelvic internal bleeding**, the engineering target should probably not be a free-cavity filler. The literature on hemorrhagic pelvic fracture repeatedly points to **transcatheter arterial embolization** as a first-line hemostatic approach for arterial pelvic bleeding, with lower complication rates when embolization is superselective rather than nonselective. A pelvic program therefore fits better as a **catheter-delivered embolic hydrogel or particulate system** than as a junctional packing derivative. [27]

For **minimally trained users**, complexity must be minimized aggressively. XSTAT's special controls required human-factors validation for emergency responders and surgeons, and low-light user testing was part of the evaluation package. That argues for field devices that are **single-use, preloaded, no mixing, intuitive**

on first use, and tolerant of stress, gloves, blood contamination, and low light. In contrast, two-component liquid systems and light-cured adhesives are naturally better matched to hospital or advanced prehospital teams. 28



Performance metrics and preclinical program

The most important performance metric is still **time to hemostasis**, but it should not be evaluated in isolation. For a junctional product, the minimally acceptable profile is rapid control of bleeding with **major reduction in blood loss**, maintenance of hemostasis during transport or movement, and **no rebleeding on removal or handoff**. For internal products, the same core endpoints matter, but so do **adhesion strength**, **burst pressure**, **resistance to fluid washout**, and **behavior on wet dynamic tissues**. Published internal adhesive systems have reported burst-pressure behavior well above physiologic arterial levels, including approximately 290 mmHg in one widely cited hydrogel. ETHIZIA's pivotal study used hemostasis at 3 minutes without rebleeding, and TCCC practice for wound dressings still emphasizes direct pressure for at least several minutes. 29

A serious bench package should include, at minimum, **whole-blood clotting or coagulation acceleration assays**, **heparinized or diluted-coagulopathic blood testing**, **absorption capacity**, **swelling ratio**, **expansion force or pressure**, **compressive recovery**, **injectability or extrusion force**, **radiopacity**, and **tissue-adhesion testing** such as lap shear, peel, wound closure pressure, or burst pressure. Notably, FDA's special controls for the XSTAT device class explicitly call for testing of absorption capacity, swelling, mechanical properties, expansion force/pressure, radiopacity, deployment functionality, retrieval, and labeling-supported use conditions. 30

Biocompatibility and stability testing should be designed from the start rather than treated as a late regulatory add-on. FDA's 2023 ISO 10993-1 guidance frames biocompatibility as a risk-based program tied to nature and duration of body contact. For sterile devices, the package and shelf-life program should cover sterilization validation, endotoxin or pyrogenicity where relevant, package integrity, and accelerated plus real-time aging. FDA materials on sterility submissions, packaging, and shelf-life—and example 510(k) summaries such as TRAUMAGEL—show the standard expectation set around **SAL 10^-6**, ISO 10993 biocompatibility, ISO 11607-style packaging validation, and aging studies consistent with ASTM F1980-style approaches. 31

A practical staged preclinical program would start with **small-animal screening** and then move quickly to **large-animal survival models**. Small-animal work is useful for down-selection—rat or rabbit tail amputation, liver punch, femoral artery puncture, infected wound, and coagulopathy models remain common in the hydrogels and cryogels literature. Large-animal work should then separate by claim: **swine femoral or subclavian artery models** for junctional products, **swine liver or hepatoportal abdominal**

injury models for cavity or surface products, and **catheter-based pelvic models** if the goal is embolization or IR delivery. [32]

For study design, the published examples suggest a sensible rule of thumb. Early feasibility studies in pigs often use **about 6 to 12 animals per arm**, while more formal, GLP-like comparison studies often move to **around 10 animals per arm or more**. XSTAT's de novo summary describes a swine femoral model with ten XSTAT-treated animals and ten controls. ResQFoam dose-ranging abdominal studies used multiple groups in the 6 to 18 animal range, plus separate control groups, to map efficacy against injury severity and pressure-related toxicity. [33]

For a first-in-class or next-generation program, the endpoints should be **hierarchical**: immediate hemostasis, blood loss reduction, survival or physiologic stabilization through a prespecified observation window, no device-related embolization or organ compression, and either complete retrieval or predictable degradation. When possible, the testing should also include **coagulopathic conditions**, because several newer systems report preserved performance in heparinized or coagulopathy-mimicking models and that is clinically relevant to severe trauma. [34]

Safety risks and mitigation

The dominant safety risks differ by platform. For **expandable temporary fillers**, the major concerns are **failure to stop bleeding, obstruction or compression of critical structures, retained fragments, infection, adverse tissue reaction, and embolization**. FDA's XSTAT classification order lays these out directly and maps them to required mitigations: non-clinical and in vivo data, sterility validation, stability assessment, biocompatibility, human-factors testing, labeling, radiopacity, and retrieval verification. [35]

For **neck applications**, the risk picture is more severe. The fact that XSTAT is not indicated for tissues above the clavicle is a strong signal that uncontrolled expansion in the neck is currently unacceptable without a very different safety profile. Any neck-directed product would need a **pressure-limited, low-swelling, highly conformal** design, likely with a stronger adhesive contribution and a weaker bulk-expansion component than a groin/axillary device. [36]

For **internal sealants and patches**, the dominant risks are **thrombosis if used intravascularly**, **infection in contaminated fields**, **adhesions**, **compression from swelling or packed material**, and sometimes **hypersensitivity or transmissible-agent concerns** when human or animal-derived materials are used. TachoSil's current label warns against intravascular use, contaminated areas, and packing cavities or closed spaces; it also notes potential tissue adhesion and transmissible-agent risks. TISSEEL warns about life-threatening or fatal air or gas embolism when sprayed outside recommended pressure and distance settings and about use for severe brisk arterial or venous bleeding. ETHIZIA warns against intravascular use, severe pulsatile bleeding, infected wounds, and use in confined spaces where swelling could compress structures. [37]

For **expansile abdominal foams**, the most important risk is not just embolization but **pressurization injury**. The swine data for self-expanding abdominal foams showed a dose-response survival benefit, but the highest-dose foam also caused increased peak intra-abdominal pressure and increased bowel injury requiring resection. That argues for firm pressure-limiting controls, strict dosing logic, and a surgical removal pathway if this concept is pursued. [38]

For **catheter/embolization approaches**, the key complication is **ischemic injury from nonselective embolization**. Pelvic trauma literature reports major complications including gluteal or skin necrosis, wound breakdown, and deep infection, with lower complication rates when embolization is more selective. A trauma embolic hemostat therefore needs not only effective occlusion but also predictable localization and controlled persistence. [39]

Thermal injury is a real historical lesson. Earlier zeolite-based QuikClot formulations were associated with painful exothermic reactions in trauma case series, and later reformulations specifically sought to reduce heat of adsorption and eliminate burn risk. Any novel chemistry that cures or crosslinks in situ should therefore be screened explicitly for **temperature rise**, not just adhesion and hemostasis. [40]

The mitigation framework is therefore straightforward. Use **radiopaque components** and retrieval counts for temporary fillers; cap **swelling and expansion pressure**; avoid **intravascular indication drift** unless the system is explicitly an embolic; validate use in **dirty blood and contaminated environments**; and design the user workflow so the safety margin depends on the device, not on heroic technique. [41]

Regulatory, manufacturing, IP, and competitor landscape

Regulatory pathways

In the United States, the pathway depends strongly on the product archetype. **Topical external hemostatic wound dressings without thrombin** currently continue to clear through **510(k)** under product code **QSY**, which remains listed as **unclassified** in FDA's product-classification database as of 2026. FDA convened an advisory panel in 2022 and stated that Class II with special controls would be appropriate, but the device type has not fully transitioned in the public database, and 510(k) clearances under QSY are still occurring in 2026. That means a new external dressing or gel for compressible or accessible bleeding may still have a predicate-based path if it fits the technology and indications of existing products. [42]

Expandable temporary internal-use junctional sponges follow a different route. XSTAT created its own de novo class-II category under **21 CFR 878.4452** for a **non-absorbable, expandable, hemostatic sponge for temporary internal use**. Future products sufficiently similar to this device type are likely to encounter a 510(k)-plus-special-controls path anchored to the XSTAT device class, assuming the intended use remains within the same risk envelope. [35]

Absorbable internal hemostatic agents and dressings are a more demanding category. FDA product codes for **LMF**, **LMG**, and related absorbable hemostatic agents are associated with **21 CFR 878.4490** and **class III/PMA** review, and products such as FLOSEAL, SURGICEL-family hemostats, Arista AH, and ETHIZIA illustrate that internal absorbable devices frequently live in PMA territory. [43]

Fibrin sealants and fibrin sealant patches in the U.S. are generally **licensed biologics** through CBER rather than ordinary device 510(k)s. TISSEEL, TachoSil, EVICEL, EVARREST, and VISTASEAL are all documented through BLA pathways or approved-blood-product listings. A product that incorporates thrombin/fibrinogen or otherwise depends on blood-derived biologic actives should therefore be expected to face a combination-product or biologics-style development burden rather than a simple device review. [44]

In the European Union, the classification logic is dictated by **EU MDR 2017/745**. Surgically invasive transient-use devices are generally class IIa under Rule 6, but if they are mainly absorbed or have biological effect the class can rise; short-term surgically invasive devices that are mainly absorbed are class III under Rule 7; implantable or long-term surgically invasive absorbable devices are class III under Rule 8; devices incorporating a medicinal substance with ancillary action are class III under Rule 14; and devices manufactured from non-viable animal or human tissue derivatives are class III under Rule 18. MDR also ties classification to clinical evaluation, PMCF, and post-market surveillance requirements. [45]

Manufacturing, sterilization, and shelf-life

Manufacturing feasibility should be treated as part of platform selection, not a later operational detail. Dry external dressings and gels already show straightforward terminal-sterilization patterns: Celox Gauze PRO and TRAUMAGEL both report terminal gamma irradiation to SAL 10^-6, and TRAUMAGEL's 510(k) summary reports a proposed 12-month shelf life. XSTAT's original de novo package established a shorter 6-month shelf life supported by real-time and accelerated aging. Those examples illustrate that battlefield-friendly products are possible, but also that shelf-life and stability are not automatic for complex deployable systems. [46]

For internal products, manufacturing becomes more complex as soon as the platform uses **reactive multi-component chemistries**, **protein biologics**, or **animal/human-derived matrices**. That complexity increases the burden of raw-material control, viral or immunogenicity assessments, and packaging configuration. FDA guidance and recognized standards make clear that sterility, biocompatibility, packaging integrity, and shelf-life all need formal validation packages rather than ad hoc demonstrations. [47]

As a commercial inference from the compositions and regulatory precedents, **dry polysaccharide/chitosan/kaolin field products** are the most likely to yield the lowest cost of goods and simplest packaging, while **plasma-derived fibrin sealants** and **multi-component reactive internal sealants** are likely to be costlier and operationally more complex. That inference is supported by the clear split between simple 510(k)-cleared external dressings and PMA/BLA-governed absorbable hemostats and fibrin products. [48]

IP considerations

The IP landscape is already crowded in the most relevant white spaces. Representative public materials point to at least four dense clusters: **expandable wound-filler applicators and sponges** around RevMedx/XSTAT; **abdominal hemorrhage control foams** and in-situ forming cavity-control devices around Arsenal Medical/ResQFoam; **catechol-chitosan or catechol-biopolymer adhesive hydrogels**; and **PEG-based crosslinked hydrogel sealants plus sterilization/tissue-sealing methods**. A new entrant should expect freedom-to-operate pressure not only on composition, but on **delivery geometry, applicator design, radiopaque markings, removability, and setting-control methods**. This is enough to justify an early formal FTO review before locking the chemistry. [49]

Comparison table of the current product landscape

Product	U.S. status	Mechanism	Best-aligned use-case	Major limitation for a new “universal” product concept
QuikClot Hemostatic Dressing / Combat Gauze	510(k) K123387; external topical hemostatic dressing [18]	Kaolin on gauze; topical procoagulant dressing	Packed external traumatic or surgical bleeding	Requires packing and pressure; not a solution for deep non-packable tracts or internal torso bleeding
Celox Gauze PRO	510(k) K113560 [50]	Chitosan-coated gauze; barrier plus platelet-concentrating effect	Moderate-severe external bleeding	External use only; still depends on packing and access
ChitoGauze	510(k) K092357 / related variants [51]	Chitosan hemostatic gauze	Severe external bleeding	Same structural limitation as other gauze products
XSTAT	De Novo K130218; class II, 21 CFR 878.4452 [36]	Injectable expanding radiopaque cellulose sponges	Deep junctional groin or axillary wounds	Not indicated for abdomen, retroperitoneum, or tissues above clavicle; requires surgical retrieval
TRAUMAGEL	510(k) K240713 [52]	Chitosan granules suspended in sodium alginate gel	Moderate-severe external bleeding where a gel is easier than gauze	External only; not an internal or cavity product
FLOSEAL	PMA P990009; absorbable collagen hemostatic agent with thrombin [53]	Flowable gelatin-thrombin matrix	Operative internal bleeding control	OR-focused; not for field deployment; intravascular and swelling risks matter
Arista AH	PMA P050038; non-collagen absorbable hemostat [54]	Microporous polysaccharide particles	Diffuse surgical bleeding	Less suited to high-pressure focal arterial bleeding; operative adjunct rather than trauma bridge

Product	U.S. status	Mechanism	Best-aligned use-case	Major limitation for a new “universal” product concept
TachoSil	BLA 125351; current U.S. label maintained through 2025 supplement [55]	Equine collagen sponge coated with human fibrinogen and thrombin	Cardiovascular and hepatic surgery	Not for major arterial/venous bleeding; not for intravascular use; avoid packing cavities
EVARREST	BLA 125392 [56]	Fibrin sealant patch	General surgical bleeding requiring compression	Manual compression required; not a substitute for ligation in major arterial/venous bleeding
ETHIZIA	PMA P240035; FDA approved 2025; CE-marked and marketed outside the U.S. from 2024 [57]	Porcine gelatin carrier with synthetic NHS-POx/NU-POx polymeric additive	Visible open liver-surgery bleeding	Indicated for minimal-moderate bleeding only; not severe pulsatile bleeding; confined-space swelling concerns
TISSEEL / VISTASEAL / EVICEL	BLAs / approved blood products [58]	Two-component fibrin sealants	Topical operative sealing and adjunctive hemostasis	Spray-use embolism risk if misapplied; not a prehospital cavity filler
ResQFoam	Investigational; company reports Breakthrough Device designation and early clinical evaluation [59]	In-situ forming self-expanding abdominal foam	Severe intra-abdominal hemorrhage as bridge to surgery	Not approved; pressurization and bowel-injury issues remain central translational hurdles

Strategic recommendation and key sources

The strongest development thesis is a **sequenced portfolio**, not a single omnibus product claim.

The **recommended first program** is a **temporary, radiopaque, pressure-limited, retrievable junctional sponge or cryogel** for groin and axillary wounds. The optimal chemistry is likely a **macroporous shape-memory scaffold** based on **cellulose, chitosan, gelatin, or recombinant collagen**, strengthened with **PEG or dopamine/catechol chemistry**, and optionally augmented with **kaolin or thrombin** if the added regulatory cost is justified. Its defining value proposition should be **easy responder deployment in deep tracts, faster application than gauze, performance in coagulopathic blood, and safe retrieval**. That is the clearest unmet need with the cleanest translational logic. [60]

The **recommended second program** is an **internal-use surgeon/IR platform**, not a neck extension of the same product. For the OR, a **wet-tissue adhesive absorbable patch or flowable hydrogel** is more credible than a bulky filler. For pelvic arterial control, a **catheter-deliverable embolic hydrogel** is more credible than a free-cavity expansion device. If the goal is to serve the neck, that should be a **third program** with much tighter swelling and compression controls than a groin/axilla product. [61]

The most important **no-go decision** is this: do **not** try to carry an abdominal-cavity or pelvic-internal claim using a product architecture whose primary mechanism is uncontrolled bulk expansion. The swine foam literature shows why. Similarly, do **not** assume that a groin/axilla filler can simply be repurposed for the neck; current regulatory precedent points the other way. [62]

The most important **open questions** are not scientific first-order questions but translational ones. These include the exact pressure window that is safe in deep cervical and junctional anatomy, whether a company wants a **temporary retrievable** versus **absorbable** product for the first program, how aggressively to incorporate **biologic actives** versus staying in a more device-like chemistry space, and the outcome of a full **freedom-to-operate** review in the expandable-sponge and adhesive-hydrogel patent clusters. Those questions can be answered, but they should be answered early because they will determine both the regulatory path and the economics. [63]

Key sources used most heavily in this report include the FDA de novo order and summary for **XSTAT**; FDA 510(k) summaries for **QuikClot**, **Celox Gauze PRO**, and **TRAUMAGEL**; FDA labels and approved-blood-product pages for **TachoSil**, **TISSEEL**, **VISTASEAL**, **EVICEL**, and **EVARREST**; the FDA SSED for **ETHIZIA**; FDA classification materials for topical hemostatic wound dressings and absorbable hemostatic agents; the EU MDR text in **Regulation (EU) 2017/745**; and peer-reviewed reviews and primary articles on **non-compressible torso hemorrhage**, **adhesive hydrogels**, **cryogels**, **foams**, and **TXA-enabled systems**. [64]

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https://pubmed.ncbi.nlm.nih.gov/39559807/

[4] [12] https://pubmed.ncbi.nlm.nih.gov/40558744/
https://pubmed.ncbi.nlm.nih.gov/40558744/

[5] [38] [62] https://pubmed.ncbi.nlm.nih.gov/24553527/
https://pubmed.ncbi.nlm.nih.gov/24553527/

[6] https://jts.health.mil/assets/docs/cpgs/Damage_Control_Resuscitation_12_Jul_2019_ID18.pdf
https://jts.health.mil/assets/docs/cpgs/Damage_Control_Resuscitation_12_Jul_2019_ID18.pdf

[11] [18] https://www.accessdata.fda.gov/cdrh_docs/pdf12/K123387.pdf
https://www.accessdata.fda.gov/cdrh_docs/pdf12/K123387.pdf

[13] [29] https://www.nature.com/articles/s41467-019-10004-7
https://www.nature.com/articles/s41467-019-10004-7

[14] [19] https://pubmed.ncbi.nlm.nih.gov/35436591/
https://pubmed.ncbi.nlm.nih.gov/35436591/

[15] https://pubmed.ncbi.nlm.nih.gov/38253524/
https://pubmed.ncbi.nlm.nih.gov/38253524/

[16] [24] [32] [33] [60] https://www.accessdata.fda.gov/cdrh_docs/reviews/k130218.pdf
https://www.accessdata.fda.gov/cdrh_docs/reviews/k130218.pdf

[17] [46] [50] https://www.accessdata.fda.gov/cdrh_docs/pdf11/k113560.pdf
https://www.accessdata.fda.gov/cdrh_docs/pdf11/k113560.pdf

[20] https://pubmed.ncbi.nlm.nih.gov/40605112/
https://pubmed.ncbi.nlm.nih.gov/40605112/

[21] https://pubmed.ncbi.nlm.nih.gov/41611431/
https://pubmed.ncbi.nlm.nih.gov/41611431/

[22] https://pubmed.ncbi.nlm.nih.gov/35200490/
https://pubmed.ncbi.nlm.nih.gov/35200490/

[23] [25] [37] [55] https://www.fda.gov/media/78698/download
https://www.fda.gov/media/78698/download

[26] [44] [58] https://www.fda.gov/vaccines-blood-biologics/approved-blood-products/tisseel
https://www.fda.gov/vaccines-blood-biologics/approved-blood-products/tisseel

[31] [47] https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-international-standard-iso-10993-1-biological-evaluation-medical-devices-part-1-evaluation-and
https://www.fda.gov/regulatory-information/search-fda-guidance-documents/use-international-standard-iso-10993-1-biological-evaluation-medical-devices-part-1-evaluation-and

[34] https://pubmed.ncbi.nlm.nih.gov/41127547/
https://pubmed.ncbi.nlm.nih.gov/41127547/

[39] https://pubmed.ncbi.nlm.nih.gov/27544681/
https://pubmed.ncbi.nlm.nih.gov/27544681/

[40] https://pubmed.ncbi.nlm.nih.gov/18404080/
https://pubmed.ncbi.nlm.nih.gov/18404080/

[42] https://www.fda.gov/media/162531/download
https://www.fda.gov/media/162531/download

[43] https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpcd/classification.cfm?id=LMF
https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpcd/classification.cfm?id=LMF

[45] https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX%3A32017R0745
https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX%3A32017R0745

[48] [52] https://www.accessdata.fda.gov/cdrh_docs/pdf24/K240713.pdf
https://www.accessdata.fda.gov/cdrh_docs/pdf24/K240713.pdf

[49] https://revmedx.com/pages/patents
https://revmedx.com/pages/patents

[51] https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?id=K092357
https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?id=K092357

[53] https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P990009
https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpma/pma.cfm?id=P990009

[54] https://www.accessdata.fda.gov/cdrh_docs/pdf5/P050038b.pdf
https://www.accessdata.fda.gov/cdrh_docs/pdf5/P050038b.pdf

[56] https://www.fda.gov/vaccines-blood-biologics/approved-blood-products/evarrest-fibrin-sealant-patch
https://www.fda.gov/vaccines-blood-biologics/approved-blood-products/evarrest-fibrin-sealant-patch

[57] https://www.accessdata.fda.gov/cdrh_docs/pdf24/P240035B.pdf
https://www.accessdata.fda.gov/cdrh_docs/pdf24/P240035B.pdf

[59] https://arsenalmedical.com/products/resqfoam/
https://arsenalmedical.com/products/resqfoam/